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Machine Learning-Based Identification of Stress Behaviours in Zoo Animals: The Role of Feeding Schedules and Keeper Routines

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ABSTRACT Pacing and swaying are common welfare indicators in zoos, but automated monitoring systems focus on movement detected by cameras and do not take into account husbandry activities that can trigger such behaviour. At the National Zoological Gardens, Dehiwala, Sri Lanka, we tested whether feeding-schedule and keeper-routine variables improve the detection of stress-indicative behaviour in two species. We recorded 190 animal sessions (96 Asian elephant, 94 toque macaque; 32.1% stress-present) over 28 observation days, using scan-sampled observations and keeper and feeding logs. A Random Forest trained on movement and context features (baseline) was compared with models that also include seven feeding and keeper-routine features, using stratified, day-grouped five-fold cross-validation repeated ten times and a held-out set of four days. The proposed Random Forest reached F1 = 0.68 $\pm$ 0.15 (AUC 0.84 $\pm$ 0.08) against 0.33 $\pm$ 0.09 (AUC 0.54 $\pm$ 0.09) for the baseline, and was better in all ten repeats (Nadeau–Bengio corrected $p < 0.001$). One feature, time to keeper arrival, accounted for nearly all of the gain, and the model missed 37% of stress sessions, most of which occurred well away from keeper arrival. The improvement was not statistically significant on the 28-session held-out set (McNemar $p = 0.51$). The data come from a single zoo, two species and one annotator, with no measure of inter-annotator agreement, so the results show a promising, keeper-actionable signal, not a deployable welfare tool. All code and anonymised data are public.

INDEX TERMS Animal welfare, feeding schedule, keeper routine, machine learning, random forest, Sri Lanka, stereotypic behaviour, zoo animals

I. INTRODUCTION

MODERN zoos should evaluate the welfare of all the animals in their care, and behaviour is the most widely used welfare indicator [1], [2]. Manual observation is slow and subjective, and machine learning is increasingly used to automate it, typically with cameras and wearable sensors that determine where an animal is and how much it moves [1], [3]–[5]. A systematic review found that only 19 studies had used technology for real-time welfare monitoring in zoos, that algorithms featured in only two of them, and that the work concentrated on large mammals [1].

In this field, stereotypic behaviour (repetitive pacing or swaying) is a common indicator of poor welfare. Husbandry practices appear to affect it, and experimental data support this. Unpredictable or random feeding reduced pacing in Asiatic lions and Amur tigers [6], [7], and cats on a three-day feeding cycle paced differently from cats fed daily [8]. Such studies, however, each tested one routine at a time and examined group averages; they did not ask whether routine variables can predict stress-indicative behaviour in individual animals. On the other hand, the automated systems listed above predict behaviour from movement but do not use schedule information [3]–[5].

Research gap. We were not able to identify any study that uses feeding-schedule and keeper-routine variables as predictors in a supervised classifier of stress-indicative behaviour, nor any such study from a South Asian zoo. The region matters: endangered Asian elephants and the endemic toque

macaque are kept in Sri Lanka's National Zoological Gardens in Dehiwala, where welfare monitoring is manual. Feeding times have been shown to affect the behaviour of leopards in Indian zoos [9], which indicates that it is worthwhile to test feeding-time variables in the region. A keeper routine is also inexpensive to record and can be altered by management, so any predictive signal is immediately actionable.

Research question. When a classifier is trained on variables describing the feeding schedule and keeper routine, is it more successful at detecting stress-indicative behaviour in zoo animals than a baseline classifier trained on general movement features, on days not seen during training? The hypothesis is testable: it is rejected if the difference in F1 is not significant at $\alpha = 0.05$ under day-grouped validation.

Approach. We analysed primary data gathered from Asian elephants and toque macaques at the Dehiwala zoo by comparing a Random Forest baseline (movement and context features) with five classifiers that add seven feeding and keeper-routine features. We evaluated them with stratified, day-grouped, repeated cross-validation, used a held-out test set, and carried out an ablation study and an error analysis.

Contributions. We make five claims, which can be tested with our public code and data:

1) When feeding and keeper-routine features are added to the movement and context baseline, F1 increases from 0.33 to 0.68 under day-grouped cross-validation.
2) A Random Forest outperforms SVM, logistic regression and a neural network on this feature set (fixed, untuned hyperparameters).
3) Almost all of the gain is due to one feature, time to keeper arrival.
4) The model misses 37% of stress sessions, mainly those far from keeper arrival.
5) The improvement is not statistically significant on the 28-session held-out set, so the evidence is cross-validation-based and preliminary.

The rest of the paper briefly reviews the relevant literature (Section II), explains the data and methodology (Section III), presents the results (Section IV), discusses them (Section V), and concludes (Section VI), commenting on the results as well as on limitations and ethical considerations.

II. RELATED WORK

A. FEEDING REGIME AND STEREOTYPIC BEHAVIOUR

Early scan-sampling studies of nine captive felid species showed that stereotypic pacing was influenced by feeding regime: cats on a three-day cycle tended to pace more on fast days, and daily-fed cats tended to pace more in the hour before feeding, although not significantly [8]. When Amur tigers were fed with electrically controlled feeding boxes at semi-random times, pacing fell from up to 16% of observed time under conventional feeding to near zero, which supports a frustrated-foraging hypothesis [7]. More recently, random feeding days were found to decrease mean pacing in Asiatic lions compared with fixed feeding days [6]. The picture is not uniform. In four South Indian zoos, feeding time affected the behavioural repertoire of leopards (less resting before feeding) but did not significantly affect stereotypic pacing, which was related more to enclosure characteristics [9]. Automatic timed feeders had no significant effect on stereotypy in one geriatric male Asian elephant observed overnight [10]. Taken together, these studies show that routine timing can matter for some species, but effects vary by species, context and measure, and all of the studies treated routine variables as experimental treatments rather than as predictors.

B. AUTOMATED MONITORING IN ZOOS

In a systematic review of technologies for monitoring zoo animals, the most common tools were cameras (52.6%) and wearable sensors (31.6%), mainly used to measure behaviour, and algorithms were used in just two publications [1]. A deep-learning framework for polar bears localised and identified individuals with an F1 of 86.4% and derived trajectories and activity from video [3]; a follow-up study used computer vision and machine learning to detect stereotypical movement over the long term and its seasonal variation [4]. PantherAI analysed CCTV footage of a zoo tiger to generate space-use maps and activity budgets for behaviours that include stereotypical pacing [5]. Smart sensing for animal welfare has also been studied in human–computer interaction [11]. These systems have one thing in common: they predict behaviour from the animal's position and movement. To our knowledge, they do not include keeper-routine timing as an explicit input, which is the gap we address.

C. MACHINE-LEARNING CLASSIFIERS FOR ANIMAL BEHAVIOUR

Random Forests [12] and support vector machines are typical classifiers for behaviour classification. In training conditions, Random Forest and SVM reached Cohen's kappa of 0.82 and 0.78 on accelerometer data from captive red foxes, but they did not perform well when transferred to wild foxes, whereas a neural network with a moving window transferred better [13]. For captive cheetahs, six classifiers and two ensembles identified feeding with a precision of 74.4–81.6% and a recall of 54.8–82.4%, with the SVM best for feeding, and accuracy fell when models were tested on animals they had not seen [14]. For sequential data, the LSTM-FCN architecture combines a recurrent (LSTM) branch with a convolutional branch for feature extraction [15]. We planned to use it for sequences of meal and keeper events but, as Section III-H explains, our data are session-level and not sequences.

D. RELIABILITY OF MANUAL OBSERVATION

Observer bias, scoring what one expects to see, is recognised as a threat in animal-behaviour studies [16]. A review of 645 live reliability tests with 163 observers found that observers were tested on only about 25% of the behaviours in the ethogram [2]. The usual remedy is chance-corrected agreement such as Cohen's kappa [17], and we return to it as a limitation of our own labels.

TABLE 1: Dataset summary by species.

Species	Individuals	Enclosures	Sessions	Stress-present
Asian elephant	6	2	96	23 (24.0)
Toque macaque	6	2	94	38 (40.4)
Total	12	4	190	61 (32.1)

III. METHODOLOGY

A. STUDY SITE, ANIMALS AND ETHICS

Asian elephants (Elephas maximus) and toque macaques (Macaca sinica) were observed at the National Zoological Gardens, Dehiwala, Sri Lanka. The dataset covers 12 individuals (six per species) in four enclosures (E1 and E2 for elephants, M1 and M2 for macaques). Observations were conducted on 28 days (alternate days) from 1 April to 25 May 2026, in both the morning and the afternoon. Permission was obtained from the zoo management. The researchers did not approach, feed or interact with any animal; they observed from public viewing areas or from areas approved by the keepers. No physiological samples were taken. Keeper data were recorded under rotating codes, not keeper names. Asian elephants are endangered, so individual animal identifiers have been removed from the public dataset and results are reported at species level.

B. DATA COLLECTION

Two researchers used instantaneous scan sampling [18] to record behaviour at 5-minute intervals in each observation session, with an ethogram derived from the zoo stereotypy literature to identify stress-indicative behaviours. The zoo-assigned keeper completed a structured daily log for each observed enclosure recording keeper arrival and departure times, meal start and end times, and diet items. The behavioural and keeper logs were merged on enclosure, date and session to create one record per animal session.

C. DATASET AND ANNOTATION

The final table has 190 records, one per animal session, from 28 observation days with 6–7 sessions per day. Table 1 gives the class distribution. Overall, 61 sessions (32.1%) are stress-present; this is higher than the 10–20% we expected for macaques from previous literature (24.0% in elephants). A session is labelled Stress_Present if stress-indicative behaviour from the ethogram was observed in that session, and Stress_Absent if it was not. Most sessions were coded by one researcher. Our protocol included an analysis of the agreement between two independent coders (Cohen's kappa [17]); this was not performed, so inter-annotator agreement is unknown, and we discuss the implications in Section V.

D. PREPROCESSING AND FEATURES

Records were validated (required columns, label values, time formats and duplicates). No record had a missing label or key value, so all 190 records were retained and no values were imputed. Clock times were converted to minutes after midnight, and three durations were computed: keeper presence (departure time minus arrival time), meal duration, and the interval from keeper arrival to meal start. In 21 sessions the meal ended after the keeper had left; these were retained and flagged. Numeric and categorical features were processed inside the modelling pipeline, with numeric features z-scored and categorical features one-hot encoded, so both were fitted on the training folds only. The features fall into four groups:

- **Movement** (3): activity level (an integer score from the observation sheet, range 25–95), location changes (count) and boundary time (minutes spent at the edge of the enclosure).
- **Context** (3): species, enclosure and session (morning or afternoon).
- **Feeding and keeper routine** (7): inter-meal interval, diet variety score, keeper proximity duration, time to keeper arrival, keeper presence duration, meal duration and keeper-arrival-to-meal interval. Time to keeper arrival is the time in minutes from the start of the observation session to the arrival of the keeper (range 0–30).
- **Extended** (1): rolling stereotypy rate. It is derived from behaviour and may overlap with the labelling, so it is analysed only as an extension and not as part of the main model.

TABLE 2: Feature means ± standard deviations by class (all 190 sessions).

Feature	Absent (n=129)	Present (n=61)
Time to keeper arrival (min)	17.2 ± 7.7	9.9 ± 9.7
Inter-meal interval (min)	239.9 ± 53.7	218.3 ± 60.2
Diet variety score	3.79 ± 0.99	3.61 ± 1.13
Keeper proximity (min)	14.4 ± 5.3	13.7 ± 5.0
Activity level	59.8 ± 17.5	60.0 ± 17.2
Location changes	5.7 ± 3.1	6.6 ± 3.5
Boundary time (min)	5.5 ± 4.3	7.7 ± 4.6
Rolling stereotypy rate	0.20 ± 0.11	0.23 ± 0.13

Table 2 summarises the main features by class.

E. SYSTEM ARCHITECTURE

Fig. 1 shows the end-to-end system. Raw keeper, observation and metadata logs are validated and merged, transformed into features, split by observation day, passed to the baseline and proposed classifiers, and evaluated. The output is a per-session stress probability and a feature-importance report for keepers and researchers.

F. BASELINE AND MODELS

The **baseline** is a Random Forest trained on the movement and context features only. Its inputs are similar to those of existing zoo monitoring systems, which are based on movement and trajectories [3], [4], and it receives the same context columns as the proposed model, so it differs from the proposed model only in the feeding and keeper-routine features. A movement-only Random Forest is also reported.

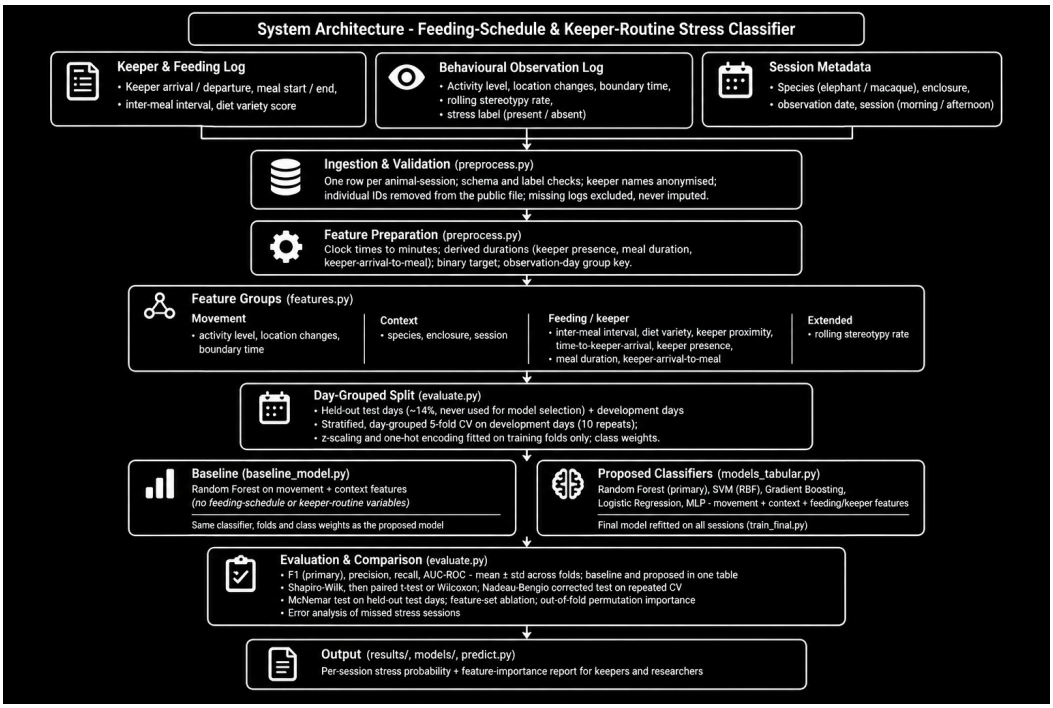


FIGURE 1: System architecture of the feeding-schedule and keeper-routine stress classifier, from raw logs to evaluated predictions.

The **proposed** models add the seven feeding and keeper-routine features. We compare five classifiers. The main model is a Random Forest [12] (300 trees, minimum leaf size 3), which copes with small, noisy, mixed-type tabular data and provides feature importances that keepers can examine. The second tabular model is an SVM with an RBF kernel, a popular choice for behaviour classification [13], [14]. Gradient boosting and logistic regression are added as further comparators. The neural model is a multilayer perceptron (two hidden layers of 32 and 16 units, L2 penalty 0.01, Adam solver, the library default). All models use class weights for the minority class where supported. We did not use SMOTE [19]: with only 162 development sessions, oversampling adds little real information and makes leakage harder to control, so we chose class weights over synthetic oversampling. Hyperparameters were fixed in advance and not tuned.

G. VALIDATION AND METRICS

Observations from the same day share the keeper's schedule and conditions, so splitting by row would allow the model to exploit similarity between observations from the same day. All splits are therefore by observation day. One of seven stratified, day-grouped folds (about 14%; four days, 28 sessions) is set aside as a final test set and is not used for model selection. The dataset is small and we wanted a large training set, so we used this 7-fold design instead of the more conventional 70:15:15 split. The remaining 162 sessions (24 days) are evaluated with stratified, day-grouped five-fold cross-validation, repeated ten times with different fold shuffles to reduce dependence on a single partition; this gives about 32 test sessions and 130 training sessions per fold.

Metrics. Stress-present is the minority class, so accuracy would be misleading; the primary metric is F1, which balances precision and recall. Precision and recall are also reported separately, because failing to detect a stress episode costs more than a false alarm, and AUC-ROC is reported as a threshold-independent measure of separability. All results are given as the mean $\pm$ standard deviation across folds.

Statistical testing. The differences between the five fold-level F1 scores of the baseline and proposed Random Forests are subjected to a Shapiro–Wilk test [20], followed by a paired t-test or a Wilcoxon signed-rank test [21]. Because fold-level tests are optimistic [22], we also apply the Nadeau–Bengio corrected resampled t-test [23], a conservative test, to the 50 fold differences from the ten repeats. On the held-out days we use McNemar's exact test [24] on paired correct and incorrect predictions. Significance is $\alpha = 0.05$. Feature importance is permutation importance (drop in F1), computed on each held-out fold and averaged. Implementation uses scikit-learn [25].

H. DIFFERENCES FROM THE ORIGINAL PROTOCOL

Our methodology report planned about 4,000 five-minute bins, an LSTM-FCN [15], SMOTE, tuned hyperparameters and a 70:15:15 split. The collected data are 190 session-level records, so there were no sequences to window and no LSTM-FCN was trained; SMOTE and tuning were not per-

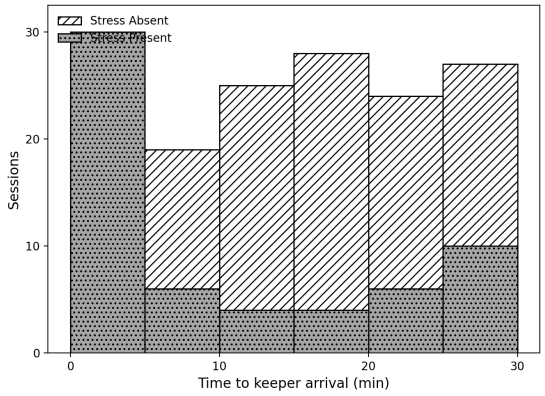


FIGURE 2: Distribution of time to keeper arrival for stress-absent and stress-present sessions.

TABLE 3: Baseline and proposed Random Forest on the same day-grouped folds. Left: ten repeats of five-fold cross-validation (mean ± SD over 50 folds). Right: single five-fold run.

Model	F1	Precision	Recall	AUC	F1 (5-fold)
Baseline (movement + context)	0.325 ± 0.093	0.343 ± 0.123	0.324 ± 0.104	0.542 ± 0.090	0.313 ± 0.072
Proposed (+ feeding/keeper)	**0.677** ± 0.147	**0.771** ± 0.121	**0.619** ± 0.179	**0.844** ± 0.082	**0.691** ± 0.084

formed, for the reasons above; and the second-coder agreement analysis was not performed. We report these deviations so that the results are not read as the experiment originally planned.

IV. RESULTS

A. DESCRIPTIVE FINDINGS

Stress-present sessions occur markedly closer to keeper arrival (mean 9.9 min) than stress-absent sessions (17.2 min) (Table 2). Fig. 2 shows that the pattern is concentrated: when the keeper arrived within 5 minutes, 85.7% of the 42 sessions were stress-present, compared with 4.3%, 17.5% and 21.0% for arrival in 5–10, 10–20 and 20–30 minutes. Movement features differ much less between the classes.

B. BASELINE VERSUS PROPOSED MODEL

Table 3 compares the baseline and the proposed model on the same folds. The proposed model's F1 is about double the baseline's (0.677 vs 0.325 over repeated cross-validation), and precision, recall and AUC are all higher. In the single five-fold run the baseline scored 0.316, 0.333, 0.200, 0.400 and 0.316 per fold and the proposed model 0.800, 0.588, 0.737, 0.632 and 0.700; it won every fold (Fig. 3). The Shapiro–Wilk test did not reject normality, so the paired t-test was used ($p = 0.0033$). The Nadeau–Bengio corrected test over the ten repeats gave $p < 0.001$ ($p = 4.8 \times 10^{-5}$), and the proposed model had the higher mean F1 in all ten repeats.

C. COMPARISON OF CLASSIFIERS

Table 4 compares the five classifiers on the proposed feature set. The Random Forest performs best on every metric.

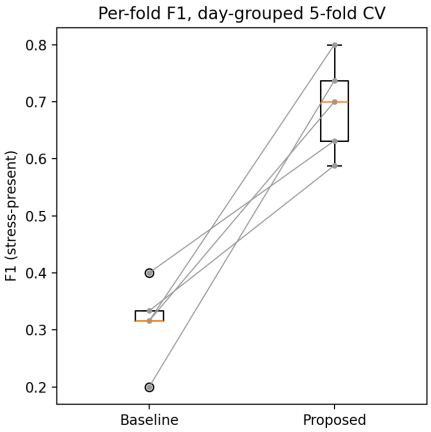


FIGURE 3: Per-fold F1 of the baseline and proposed Random Forests (five-fold, day-grouped); lines join the same fold.

TABLE 4: Classifiers on the proposed feature set (five-fold day-grouped CV, mean ± SD over folds).

Model	F1	Precision	Recall	AUC
Random Forest	**0.691** ± 0.084	**0.767** ± 0.080	**0.635** ± 0.108	**0.854** ± 0.064
Gradient boosting	0.633 ± 0.141	0.657 ± 0.161	0.613 ± 0.129	0.826 ± 0.074
SVM (RBF)	0.529 ± 0.075	0.492 ± 0.083	0.593 ± 0.139	0.691 ± 0.067
Logistic regression	0.511 ± 0.093	0.506 ± 0.187	0.558 ± 0.148	0.726 ± 0.101
MLP	0.461 ± 0.124	0.520 ± 0.145	0.425 ± 0.138	0.644 ± 0.071

Gradient boosting is close in AUC but has larger variance. The SVM, logistic regression and neural network reach F1 of 0.46–0.53, none of them far above the baseline's 0.31 in the same single-run protocol, although their hyperparameters were not tuned.

D. ABLATION AND FEATURE IMPORTANCE

Table 5 shows which features carry the signal (Random Forest, ten repeats of five-fold cross-validation). Adding species, enclosure and session to the movement features did not help (0.325 vs 0.360), so context is not the source of the improvement. Adding only time to keeper arrival to the baseline gives F1 = 0.675, essentially equal to the full proposed set (0.677). Adding the rolling stereotypy rate did not raise F1 (0.668) but slightly raised AUC (0.858).

Permutation importance (Fig. 4) agrees: shuffling time to keeper arrival lowers F1 by 0.351, shuffling inter-meal interval lowers it by 0.028, and every other feature changes it by less than 0.01 or even negatively, which indicates noise.

E. HELD-OUT TEST DAYS

On the four held-out days (28 sessions, 9 stress-present), the proposed Random Forest had F1 = 0.500, precision 1.000, recall 0.333 and AUC 0.807, against 0.471, 0.500, 0.444 and 0.538 for the baseline (Table 6). McNemar's exact test gave $p = 0.51$. This set is too small to have useful power, and the difference is therefore not statistically significant.

V. DISCUSSION

TABLE 5: Feature-set ablation (Random Forest, ten repeats of five-fold CV, mean ± SD over 50 folds).

Feature set	F1	Precision	Recall	AUC
Movement only	0.360 ± 0.097	0.381 ± 0.113	0.354 ± 0.108	0.567 ± 0.091
Baseline (movement + context)	0.325 ± 0.093	0.343 ± 0.123	0.324 ± 0.104	0.542 ± 0.090
Baseline + time to keeper arrival	0.675 ± 0.132	0.725 ± 0.123	0.650 ± 0.172	0.849 ± 0.087
Proposed (all feeding/keeper)	0.677 ± 0.147	0.771 ± 0.121	0.619 ± 0.179	0.844 ± 0.082
Proposed + stereotypy rate	0.668 ± 0.132	0.784 ± 0.107	0.601 ± 0.166	0.858 ± 0.068

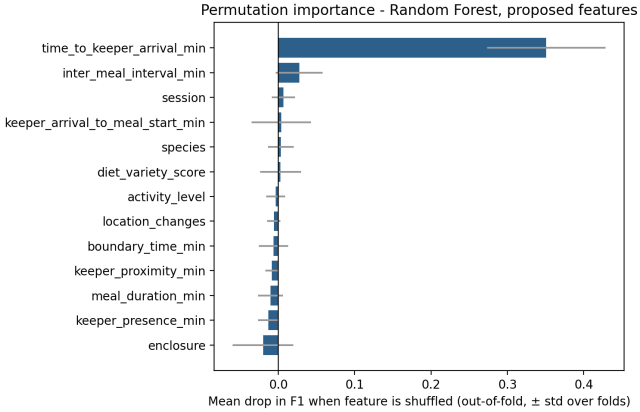


FIGURE 4: Out-of-fold permutation importance of the proposed Random Forest (mean drop in F1, ± SD over folds).

A. INTERPRETATION

Under day-grouped validation, the feeding and keeper-routine features gave a much clearer signal of stress-indicative behaviour than movement and context did. This is consistent with food-anticipatory accounts of stereotypy [6]–[8]: stress-present sessions cluster in the minutes before the keeper arrives. Our finding contrasts with the null effect of feeding time on pacing in leopards in South Indian zoos [9], which may be due to species differences, differences in how behaviour was measured (a session-level label here, time budgets there) or genuine differences between zoos. It also gives useful information for practice: the other six routine features added nothing, and the diet-variety score and meal duration were unimportant. In practice, time to keeper arrival is easy to record, and a keeper could compare it with a stress probability.

Two alternative explanations should be considered. First, the session-level label and the feature may be partly structural: sessions with an early keeper arrival contain a different mix of pre- and post-arrival time, so the strong relation might reflect how sessions were cut rather than anticipation. Finer-grained, bin-level labels would separate these. Second, if the coder knew when the keeper would arrive, expectation bias [16] could inflate the association; we did not blind the coders. Because of the strength of the effect (85.7% stress-present when arrival was within 5 minutes), both checks are worthwhile before any causal reading.

B. FAILURE ANALYSIS

(1) A leakage-prone first pipeline. In our first exploratory notebook we split rows at random, ranked seven models by accuracy and chose among them using the test set. We noticed the problem when we checked the code against our own protocol: sessions from the same day share the keeper's schedule, so a random split lets a model learn same-day context, and accuracy is a poor metric for a minority class. We rewrote the pipeline with day-grouped splits, a test set held out from model selection, F1 as the primary metric and scaling inside each training fold, and we added assertions and unit tests that no day appears in both a training and a test fold.

TABLE 6: Held-out test days (28 sessions, 9 stress-present).

Model	F1	Precision	Recall	AUC
Baseline	0.471	0.500	0.444	0.538
Proposed	0.500	1.000	0.333	0.807

TABLE 7: Missed stress-present sessions (out-of-fold, proposed Random Forest, development days).

Group	Stress sessions	Missed	Miss rate (%)
All	52	19	36.5
Asian elephant	19	6	31.6
Toque macaque	33	13	39.4
Morning	29	10	34.5
Afternoon	23	9	39.1
Enclosure E1 / E2	7 / 12	3 / 3	42.9 / 25.0
Enclosure M1 / M2	16 / 17	8 / 5	50.0 / 29.4

(2) A confounded baseline. In that first version the baseline received movement features only, whereas the proposed model also received species, enclosure and session. Any improvement could then have come from context rather than from feeding and keeper variables. We gave the baseline the same context columns and added the ablation. The correction turned out to matter little (movement-only F1 0.360 vs baseline 0.325), but it removes a confound.

(3) The dominant miss mode. We analysed out-of-fold predictions of the proposed Random Forest on the 52 stress-present development sessions (Table 7). The model missed 19 (36.5%). Correctly detected stress sessions had a mean time to keeper arrival of 3.4 minutes, whereas missed ones had 19.5 minutes: the model detects stress that occurs just before the keeper arrives and largely misses stress that occurs when no arrival is imminent. This shows that the model has learned a keeper-arrival pattern rather than a general stress signature, and it explains why movement features, which carried little signal in this dataset, could not compensate. Miss rates were higher for macaques (39.4%) than for elephants (31.6%) and highest in enclosure M1 (50.0%), although the subgroup counts are small.

(4) An unstable test-set verdict. An early exploratory run, with a slightly different split, put 27 sessions in the test set and gave a significant McNemar result ($p \approx 0.02$), whereas the four test days of the final pipeline gave $p = 0.51$. The conclusion depended on which few days were held out. We therefore treat repeated day-grouped cross-validation

as the primary evidence and report the test set as a low-power confirmatory check, rather than picking the split that favoured our hypothesis.

(5) Weak non-tree models. The SVM, logistic regression and neural network scored F1 between 0.46 and 0.53. With about 130 training sessions and untuned settings, we cannot conclude anything about the suitability of these models in general.

C. ETHICAL REFLECTION

Dataset bias and fairness. The data are from one zoo, two species and 12 animals observed by two researchers, so the model may have learned the routine of this zoo rather than a general relation. Miss rates vary by species and enclosure (Table 7), and using a single threshold for every enclosure could systematically under-detect stress in macaques or in a particular enclosure. The labels rest on human judgement with no measured inter-annotator agreement, and behaviour is merely a proxy for welfare. Observations took place only in daytime sessions, so there is no information on night-time behaviour [10].

Social impact at scale. An affordable, understandable signal might help keepers fine-tune arrival time or enrichment before stress occurs. The risks are over-reliance and false reassurance: with 37% of stress sessions missed, a "no stress" output must never replace keeper judgement. A tool that mainly identifies stress close to keeper arrival could also lead managers to change the routine cue rather than address underlying causes such as enclosure design [9]. Any intervention based on our findings should be tested with the zoo's welfare staff, preferably as a change in routine implemented as a randomised trial, as in earlier feeding studies [6], [7].

Privacy. Keeper routine data describe staff behaviour and could become a performance-monitoring instrument if used improperly. We therefore recorded no keeper names, used only session-level timing, released no keeper identifiers, and recommend that such information is used for animal welfare and not for staff appraisal. Because Asian elephants are endangered, animal identifiers have been removed from the public dataset and results are reported by species.

D. LIMITATIONS

The dataset is small (190 sessions, 28 days) and comes from a single site. The same animals appear on several days, and our day-grouping does not ensure generalisation to unseen animals. Hyperparameters were not tuned, a single coder labelled most sessions with no measured agreement, the label is at session level, and the held-out set is small. One feature dominates the effect, and its causal role is uncertain (Section V-A).

VI. CONCLUSION

We asked whether feeding schedules and keeper routines can be used to detect indicators of stress in Asian elephants and toque macaques at the Dehiwala zoo. A Random Forest using these variables significantly improved on the F1 of a movement-and-context baseline (0.68 vs 0.33, Nadeau–Bengio $p < 0.001$), but the vast majority of the improvement came from one variable: time to keeper arrival. The model missed 37% of stress sessions, mainly those away from keeper arrival, and the improvement was not statistically significant on the small held-out set. The study therefore supports the hypothesis as tested in cross-validation and leaves it open for out-of-sample confirmation.

Future studies should (i) record five-minute bin-level data and analyse them with sequence models such as LSTM-FCN [15]; (ii) include a second independent coder and report Cohen's kappa [17]; (iii) use leave-individuals-out and multi-zoo validation; and (iv) test the signal prospectively by varying keeper arrival time in a controlled study.

DATA AVAILABILITY

All code, the anonymised session-level dataset, results, the trained model and the software environment specification are available at https://github.com/n-wijerathne/stress-classifier-zoo under the MIT licence. For ethical reasons, raw zoo logs are not shared.

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